

PROPOSED METHODOLOGY

Reference-Free Adaptive Retrieval Tuning for RAG Systems Using Uncertainty and Consistency Signals

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1. Methodology Overview

The proposed work extends a standard Retrieval-Augmented Generation (RAG) pipeline with a reference-free feedback mechanism for retrieval tuning. Instead of using human-written ideal answers to judge a candidate configuration, the system estimates answer reliability using signals already available inside the RAG process. Three signals are used: retrieval confidence, semantic uncertainty, and answer-context consistency. These signals are combined into a single score, and the retrieval configuration with the highest score is selected automatically.

2. Inputs and Outputs

Input 1 - Knowledge Base: A collection of PDF or text documents, or benchmark passages, from which evidence is retrieved.

Input 2 - User Query: A natural-language question submitted to the RAG system.

Output 1 - Final Answer: The answer generated using the evidence retrieved by the best-scoring configuration.

Output 2 - Selected Configuration: The chosen chunk size, Top-K value, and embedding model.

Output 3 - Reliability Information: Retrieval confidence, semantic uncertainty, answer-context consistency, and the final combined score.

3. Overall Processing Flow

Knowledge Base + User Query

- > Create / load candidate RAG configurations
- > Retrieve Top-K chunks for each configuration
- > Generate multiple answers for each retrieved context
 - > Compute Retrieval Confidence (R)
 - > Compute Semantic Uncertainty (U)
 - > Compute Answer-Context Consistency (C)
 - > Normalize and combine the signals
- > Select the configuration with the highest reference-free score
- > Generate and return the final answer

4. Candidate Configuration Space

To keep the prototype simple and computationally practical, the first implementation uses a small search space. For example, chunk sizes of 256 and 512 tokens and Top-K values of 3 and 5 can be tested with one embedding model. After the pipeline is stable, a second embedding model can be added. This produces a compact but sufficient set of candidate configurations for demonstrating adaptive selection.

5. Proposed System Modules

Module 1 - Data Preparation and Chunking: The document collection is cleaned and divided into chunks. Separate chunked versions are created for each candidate chunk size. These versions can be prepared once and reused across queries.

Module 2 - Embedding and Vector Indexing: Each chunk is converted into a dense vector using a sentence embedding model and stored in a FAISS index. The prototype starts with a compact model such as MiniLM and can later compare a second model such as BGE-small.

Module 3 - Candidate Retrieval: The user query is embedded and compared with the stored chunk vectors. For every candidate configuration, the system retrieves the Top-K most similar chunks and records their similarity scores.

Module 4 - Response Generation: The retrieved evidence and the user question are inserted into a grounded prompt. A small instruction-tuned language model generates an answer. Three sampled answers are generated per configuration so that response uncertainty can be estimated.

Module 5 - Retrieval Confidence: Retrieval Confidence (R) measures how strongly the retrieved chunks match the query. In the basic implementation, R is the mean cosine similarity of the Top-K retrieved chunks. Higher values represent stronger retrieval confidence.

Module 6 - Semantic Uncertainty: The multiple generated answers are embedded using a fixed sentence-embedding model. Pairwise semantic similarity is calculated among the answers. Semantic Uncertainty (U) is defined as one minus the average pairwise similarity. Larger disagreement therefore produces higher uncertainty.

Module 7 - Answer-Context Consistency: A pretrained Natural Language Inference model receives the retrieved context as the premise and the generated answer as the hypothesis. The entailment probability is used as the consistency score (C). A higher value indicates stronger support from the retrieved evidence.

Module 8 - Adaptive Scoring and Selection: For the same query, R, (1-U), and C are normalized across candidate configurations. The initial reference-free objective uses equal weighting: $J = [R + (1-U) + C] / 3$. The configuration with the maximum J is selected. No ground-truth answer is used during this selection.

6. End-to-End Algorithm

1. Receive the user query and load the pre-built vector indexes for the candidate configurations.
2. For each configuration, retrieve Top-K evidence chunks and record retrieval similarity scores.
3. Generate three answers from the same retrieved context.
4. Compute R from retrieval similarity, U from semantic disagreement, and C from NLI entailment.
5. Normalize the three reliability dimensions across the candidate configurations.
6. Compute the combined reference-free score J and select the configuration with the maximum value.
7. Generate a deterministic final answer using the selected context and report the selected configuration and reliability scores.

7. Experimental Design

The evaluation clearly separates configuration selection from benchmark evaluation. During adaptive selection, the proposed method is not allowed to access the benchmark answer. Ground-truth answers are used only after the final response is generated, so that the selected configuration can be evaluated objectively without violating the reference-free design.

7.1 Systems to Compare

Fixed RAG Baseline: Uses one fixed chunk size, Top-K value, and embedding model for every query.

Proposed Adaptive RAG: Chooses the configuration using only R, U, and C, without a reference answer.

Reference-Based / ReRAG-Style Baseline: Uses similarity with the hidden ground-truth answer to identify a best configuration. This acts as a supervised comparison or upper-reference baseline, not as part of the proposed inference process.

8. Performance Metrics

Exact Match (EM): Percentage of generated answers that exactly match the benchmark answer after text normalization. Suitable for short factual QA.

Token-Level F1: Measures token overlap between the generated answer and the benchmark answer. It is more tolerant than exact match when wording differs.

Answer-Context Faithfulness / NLI Score: Measures how strongly the final answer is supported by the retrieved evidence.

Retrieval Quality: Uses retrieval similarity and, when gold evidence is available, Recall@K or Hit@K to measure whether relevant evidence was retrieved.

Configuration Selection Quality: Measures how often the configuration chosen by the reference-free objective is the same as, or close to, the best configuration according to hidden ground-truth evaluation.

Efficiency: Reports average response latency, number of LLM generations, token usage, and memory requirements. This is important because semantic uncertainty requires repeated generation.

9. Ablation Study

The contribution of each signal is evaluated by repeating the same experiment with: R only; (1-U) only; C only; R+(1-U); R+C; (1-U)+C; and the complete R+(1-U)+C objective. The aim is to determine whether combining the three complementary signals gives better or more stable answer quality than using any single signal alone.

10. Minimal Implementation Plan

- Environment: Google Colab using Python, Hugging Face Transformers, Sentence-Transformers, FAISS, and PyTorch.
- Dataset: begin with 20 SQuAD-style examples for debugging, then scale to about 100-500 questions for the final experiment.
- Initial search space: 2 chunk sizes x 2 Top-K values; add a second embedding model only after the pipeline is stable.
- Generator: one small instruction-tuned language model, such as a Phi-3 Mini or Llama-3.2-3B class model.
- NLI evaluator: a pretrained MNLI/DeBERTa-style entailment model.
- Outputs: final answer, selected configuration, signal scores, results CSV, comparison results, and ablation results.

11. Expected Outcome

The expected outcome is a compact prototype showing that retrieval settings can be selected automatically without domain-specific human-written ideal answers. The primary goal is not necessarily to outperform a supervised oracle, but to approach its answer quality while removing the annotation requirement. This directly demonstrates the central research idea: reference-free adaptive retrieval tuning using complementary retrieval, generation-uncertainty, and evidence-grounding signals.